ISAC 868 MHz LoRaWAN Antenna-Sensor — System Design & Manufacturing Summary (Rev E.5)

Hardware Revision: Rev E.5 (fab/ISAC_868_RevE5_PCBWay_Turnkey.zip)
Rule-Deck Lock SHA-256: ff1d145fc6dc1ce66f936c706389190a03c52649462288b6d8774e5a09a1cc60
Target Radio / MCU: RAKwireless RAK3172LP-SIP-8-SM-NI (STM32WLE5JC + SX1262 RFO_LP + **3.0 V** DIO3 **TCXO** **0x06** , EU868)

---

1. Executive Architecture Overview

ISAC_868_KiCad9 (Rev E.5) integrates **868 MHz LoRaWAN Class A wireless telemetry (+14.0 dBm TX with adaptive SF7/SF9/SF12 wet-radome link recovery)** and **evanescent surface-permittivity sensing (+4.0 dBm 3-tone CW reflectometry at 863.1 / 866.5 / 869.5 MHz , < 1.2 ms/tone)** onto a single 90.00 × 40.00 × 1.60 mm (+0.03/-0.10 mm) 2-layer standard FR-4 (1.51 mm core, 1 oz ENIG) pure-SMD PCBA (67 populated SMD parts, 0 THT parts) housed inside a **100% uniform-flat-roof (Z = +8.50 mm , -1.44 mm compliant skirt), open-East self-draining 0.80 mm UV-stabilized ASA/PETG** waterproof sensing-window enclosure (95.7 × 45.7 × 16.1 mm , unbroken 360° Parker ORD 5700 1.80 × 0.76 mm EPDM O-ring face-seal groove [0.92 mm gland: 23.3% squeeze, 68.3% fill] + wall-gusseted M2.5 × 22 mm EPDM-sealed corner bosses + 0 PCB-bay roof slots) bonded with **25 µm 3M 467MP** acrylic transfer adhesive (0.00 mm air gap).

1. **Full-Wave 3D Maxwell FDTD Antenna-Transducer (AE1) + Parasitic-Pre-Distorted Elliptic Harmonic Pi-Filter:**
2. Synthesized from 3D Yee-grid FDTD (rf/fdtd_3d_ant.c + rf/ifa_enclosure_mom.py , modeling exact IPC-4552 ENIG NiP/Au RF sheet resistance R_S = 23.74 mΩ/square and volumetric substrate loss): enclosed Dry Air resonance **f0 = 869.00 MHz** , raw antenna |S11|(866.5) = -23.69 dB (-20.83 dB cascaded), **all three FR-4 permittivity corners (εr ∈ {4.2, 4.4, 4.6}) satisfy |S11| ≤ -10.21 dB across 863–870 MHz** , total radiated efficiency **η_tot = 45.5%** (η_rad = 50.0%), and **43.73 MHz/RIU** Air-to-Ice resonance shift (869.0 MHz → 834.5 MHz).
3. Populated parasitic-pre-distorted 3rd-order elliptic low-pass pi-network (C_M1 = 3.6 pF GJM1555C1H3R6WB01D , L_M1 = 8.2 nH LQW15AN8N2G00D || C_MZ = 0.5 pF GJM1555C1HR50WB01D , C_M2 = 4.3 pF GJM1555C1H4R3WB01D) suppresses 2f0 (1.733 GHz) by **≥ 37.2 dB** and 3f0 (2.600 GHz) by **≥ 60.1 dB** across Dry Air and Water VSWR extremes (ETSI EN 300 220-1 ≤ -36 dBm).
4. Continuous bare-ENIG F.Mask window (X_gbr = [73.40..87.80] mm, Y_gbr = [6.65..20.80] mm , 0 silkscreen in X_gbr ≥ 50.0 mm) eliminates LPI mask ridges under the self-draining 0.80 mm flush deck.
5. **ADI LT5538IDD#TRPBF -Terminated Distributed Microstrip Return-Loss Bridge & 2-Point Calibrated Classifier:**
6. Orderable **Yageo RT0402BRD078R2L** (8.2 Ω ±0.1%) series sense resistor R1 , 0.1% Susumu RG1005P arms (R2 = 681 Ω, R3 = 3.57 kΩ, R_TERM = 56 Ω), ±2% Murata LQW15AN inductors (L_Q = 27 nH, L_S = 30 nH), ±0.05 pF GJM1555C1H capacitors (C_S = C_S2 = 1.0 pF), and ADI LT5538IDD#TRPBF (8 mm T&R) Table 1 IN+ complex load (R_par = 291.65 Ω || C_par = 1.194 pF at 866.5 MHz).
7. Compact 1.44 mm /BR_S micro-cluster over a **2.20 × 2.20 mm F.Cu + B.Cu ground cutout** (X_gbr = [33.40..35.60], Y_gbr = [13.62..15.82] mm , C_par = 0.1305 pF < 0.15 pF , distributed PCB microstrip phases included) achieves **-0.802 dB** insertion loss, **24.81 dB** 50 Ω directivity at 866.5 MHz , and **17.60 dB P01 Air-to-Ice contrast** (100.0% ≥ 6 dB yield across 2,000 Monte-Carlo trials).
8. Stepping CW sensing TX power to **+4.0 dBm** keeps LT5538IDD#TRPBF (U2) input strictly ≤ -25.40 dBm (+12.40 dB linear headroom below -13 dBm compression; unpowered +14 dBm LoRa TX keeps U2 IN+ ≤ 39.5 mV_pk << 350 mV ESD knee across VSWR 6:1), and **2-point factory calibration** (Point 1 Dry-Air Vout tare + Point 2 Al2O3 high-εr level reference cal_ref_cdb , plus 0.015 dB/°C combined thermal residual compensation) achieves **99.72% binary Water vs IPA yield (P01 separation = 0.65 dB ≥ 0.50 dB)** and **99.45% 4-hazard yield**.
9. **Zero-Dropout VBAT -Fed Detector Power Tree, Direct-Soldered J2 = DNP Battery Pads, 3.0 V TCXO, Isat = 300 mA Wirewound SMPS & Split PA8/PA9 Gating:**
10. J2 (B2B-PH-K-S) is marked **DNP = Yes** so the Turnkey PCBA is **100% single-pass automated top-side SMT (67 pure-SMD placements in ISAC_868_Turnkey_CPL_Top.csv , 0 THT headers)** and 24–26 AWG silicone battery leads solder directly upward from B.Cu into bare-ENIG Ø0.75 mm PTH pads (**Pad 1 Square = BLACK / GND , Pad 2 Round = RED / +VBAT_RAW** , 4.35 V DC MAX [NO 5V PSU], DOWSIL 3145 RTV strain relief),

eliminating the +11.50 mm roof bulge for a **100% uniform flat waterproof top cover (+8.50 mm)**; Q3 (AO3401A P-MOSFET ideal diode, < 6 mV drop), U3 (TPS7A0233DBVR 25 nA LDO, total derated C_OUT_eff ≈ 13.8 µF ≤ 22 µF), **Q1 (TPS22919DCKR + C_Q1 = 100 nF) fed directly from VBAT** (3.0–4.35 V zero-dropout supply to LT5538 , cutting U3 peak load to **24.1 mA = 8.30× margin**!), **L_SMPS = LQH2MCN150K02L** (15 µH wirewound power inductor, Isat = 300 mA), and **Q2 (NX3008CBKS,115) split onto PA9 (BAT_MON_EN , U1 Pad 50 + R_BPD = 100 kΩ)**.

11. Firmware configures DIO3 TCXO to RAKwireless-mandated **3.0 V (0x06)**, leaves PA2/PA3 in Analog High-Z (ISAC_DEBUG_UART=0), marks J2/J3/J4/J5 **DNP = Yes** (retaining bare- ENIG PTH solder/pogo-pin pads with B.SilkS legends inside the sealed enclosure), enforces LoRaWAN 1.0.3 §7 Join backoff (17.281 s Tier-3) + §5.4 RXParamSetupReq (0x05) + commit-last NVM with 72-bit Flash ECCD NMI self-healing, promotes uplinks to SF12 (DR0) under wet-radome loading, and logs per-unit UID64 + 2-point cal into provisioning_vault/provisioned_units_secret.csv (--lock-rdp1).

2. Verified Rev E.5 Key Metrics (firmware/test/data/design_values.json)

Subsystem Metric	Verified Value	Acceptance Requirement
PCB / Assembly Counts	92 footprints (67 pure-SMD pop, 0 THT; 10 DNP incl. J2 ; 15 BOARD_ONLY), 131 tented vias	0 ERC, 0 DRC, 0 DNP paste
IPC-4552 ENIG NSMD Mask & CPWG Z0	0.035 mm global / 0.025 mm U1,U2 NSMD (≥ 0.180 mm dams); Z0 = 51.90 Ω (51.2–57.1 Ω , RL ≥ 26.8 dB)	ImpedanceControlled = False
Bridge Through Loss & 50 Ω Directivity	|S21| = -0.802 dB , Directivity(866.5) = 24.81 dB (C_par = 0.1305 pF)	≥ 22.0 dB (< 0.15 pF)
2-Point Calibrated 2,000 -Trial MC Yield	Air-to-Ice **P01 = 17.60 dB** (100% yield); Water vs IPA **P01 = 0.65 dB (99.72% yield)**	P01 ≥ 8.0 dB / ≥ 0.50 dB
Enclosed AE1 Resonance, εr Corners & η_tot	f0 = 869.0 MHz , |S11|(866.5) = -20.83 dB , εr∈{4.2,4.4,4.6} worst |S11| ≤ -10.21 dB , η_tot = 45.5%	|S11| ≤ -10.0 dB all εr
Harmonic Pi-Filter Attenuation (2f0 / 3f0)	37.20 dB (1.733 GHz) / 60.08 dB (2.600 GHz) enclosed (37.46 / 60.65 dB water)	≥ 28 dB / ≥ 45 dB
Uniform Flat Waterproof Enclosure (95.7 × 45.7 × 16.1 mm)	Flat +8.50 mm roof (-1.44 mm compliant skirt) + 0.80 mm open-East ASA/PETG deck + 25 µm 3M 467MP + 1.80 × 0.76 mm Parker ORD 5700 O-ring (23.3% squeeze, 68.3% fill, M2.5 × 22 mm EPDM-sealed corner screws, 0 roof slots)	0 3D collisions (77 items)

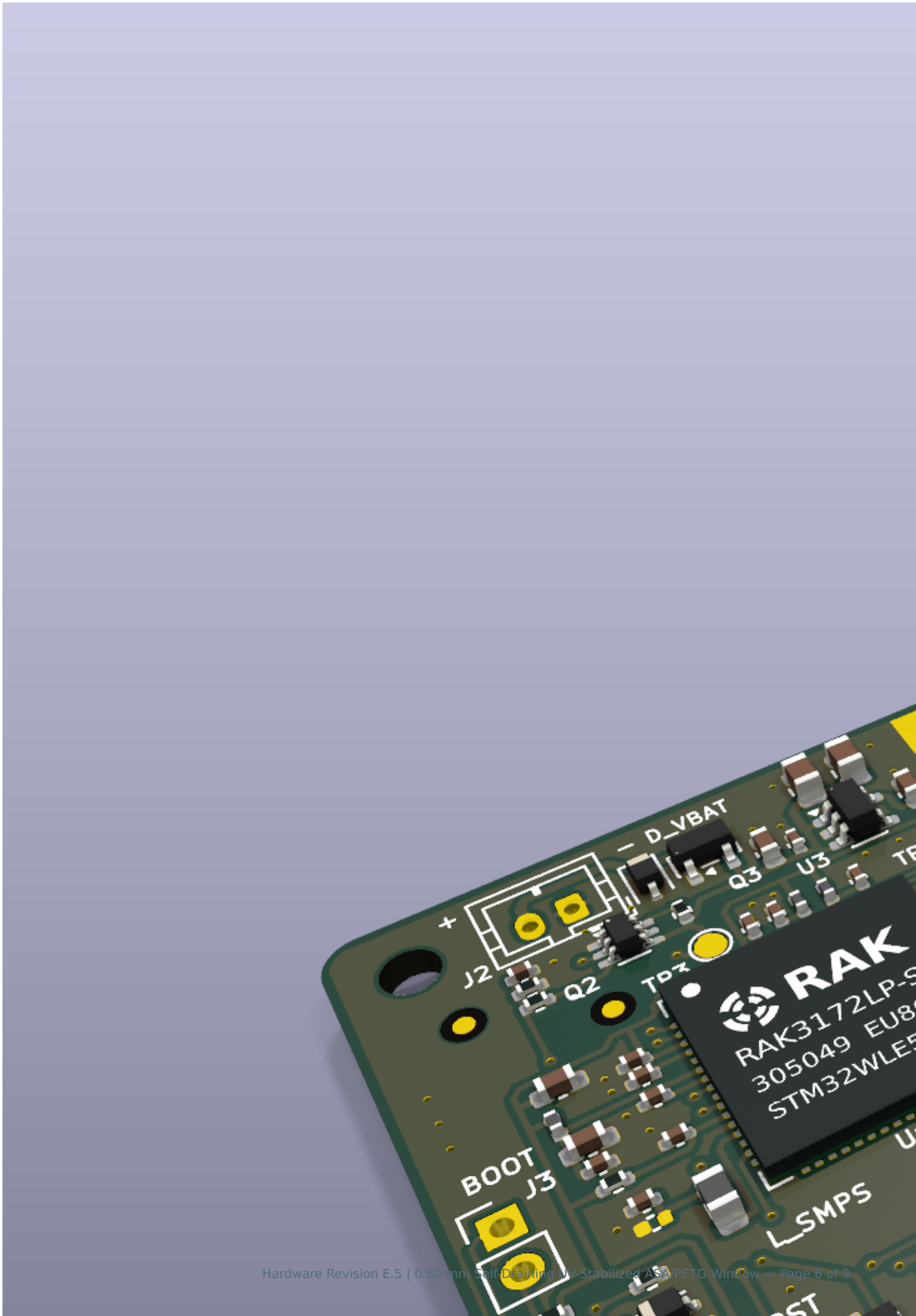
3. Full-Wave 3D FDTD 3-Tone Benchmark Table (+4.0 dBm CW Sensing, 2-Point Calibrated)

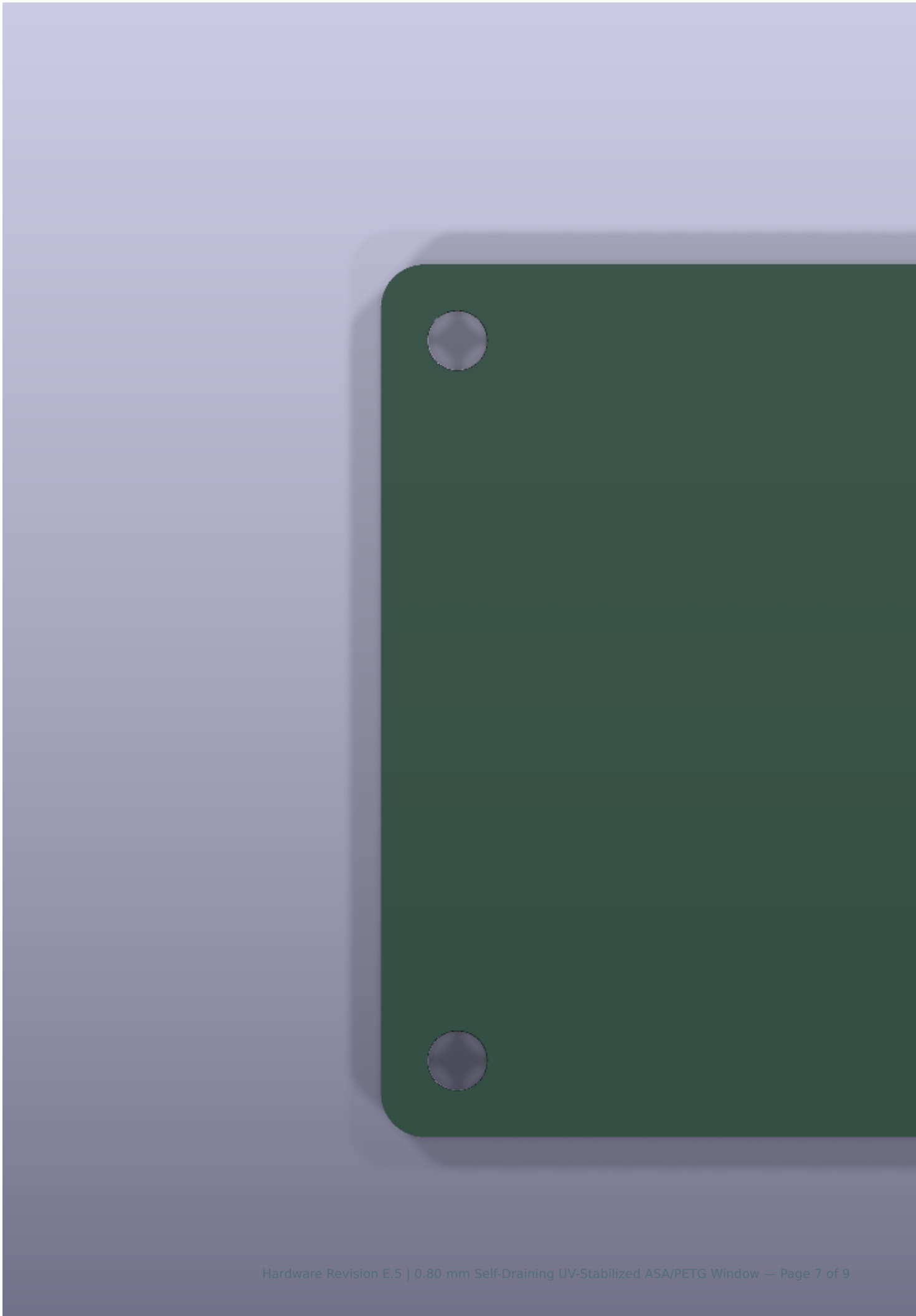
External Superstrate (2.0 mm on 0.80 mm ASA/PETG)	ε_r^* (868 MHz)	f_{res} [MHz]	Cascaded |S11| (f1 / f2 / f3) [dB]	P_det (f1 / f2 / f3) [dBm]	Tilt [dB]	Notch [dB]	Level [dBm]	FPort 10 State
Dry Air	1.00 - j0.000	869.0	-18.79 / -20.83 / -18.33	-45.75 / -48.92 / -42.31	-3.44	+4.88	-45.66	0 = DRY_AIR
PTFE / Oil	2.10 - j0.0002	849.0	-8.72 / -7.14 / -5.99	-33.83 / -33.36 / -32.60	-1.23	+0.15	-33.26	1 = OIL
Glaze Ice	3.20 - j0.003	834.5	-3.65 / -2.89 / -2.32	-28.61 / -28.66 / -28.60	-0.01	+0.06	-28.63	2 = GLAZE_ICE

External Superstrate (2.0 mm on 0.80 mm ASA/PETG)	ε_r^* (868 MHz)	f_{res} [MHz]	Cascaded |S11| (f1 / f2 / f3) [dB]	P_det (f1 / f2 / f3) [dBm]	Tilt [dB]	Notch [dB]	Level [dBm]	FPort 10 State
Alumina (Al2O3)	9.80 - j0.002	783.0	-0.36 / -0.53 / -0.69	-25.40 / -26.21 / -26.96	+1.56	+0.03	-26.19	3 = IPA (Cal Ref)
IPA (868 MHz)	6.29 - j6.090	794.0	-2.27 / -2.19 / -2.12	-27.01 / -27.77 / -28.46	+1.45	+0.04	-27.75	3 = IPA
Water Film	79.9 - j3.810	776.0	-0.71 / -0.72 / -0.71	-25.73 / -26.38 / -26.95	+1.22	+0.04	-26.35	4 = LIQUID_WATER

4. Visual Inspection Sheets (`Rev E.5` PCBA & Self-Draining Weatherproof `ASA/PETG` Enclosure)

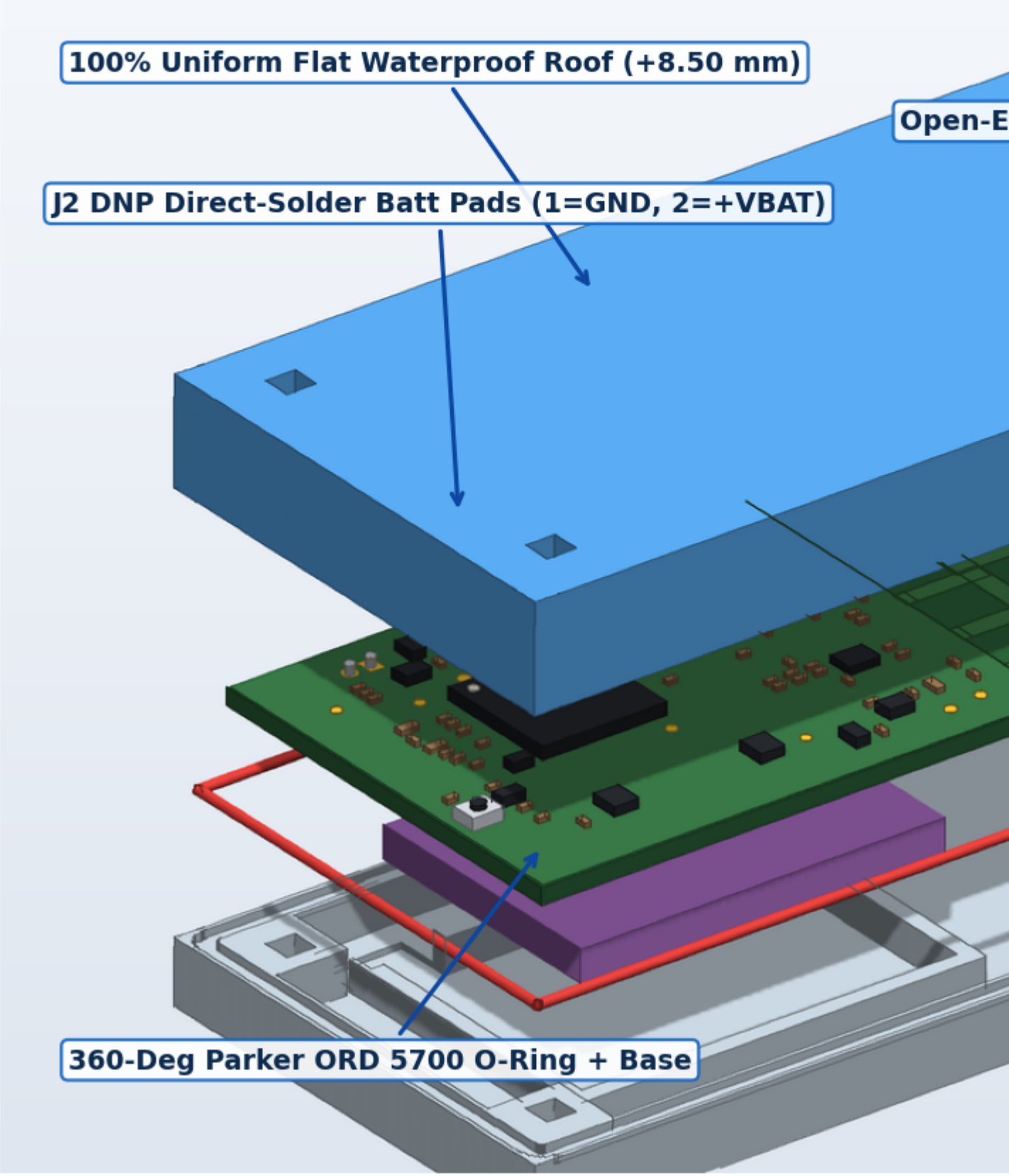

+
– D_VBAT
J2
Q3
U3
Q2
TP3
RAK
RAK3172LP-SiP
305049 EU868
STM32WLE5JC
BOOT
J3
U1
L_SMPS
D_RST
SW1
D_UA
SWD
D_SWD
J5
J4


D_VBAT
Q3
U3
J2
Q2
BOOT
J3
L_SMPS
RAK3172LP-S
305049
STM32WLE

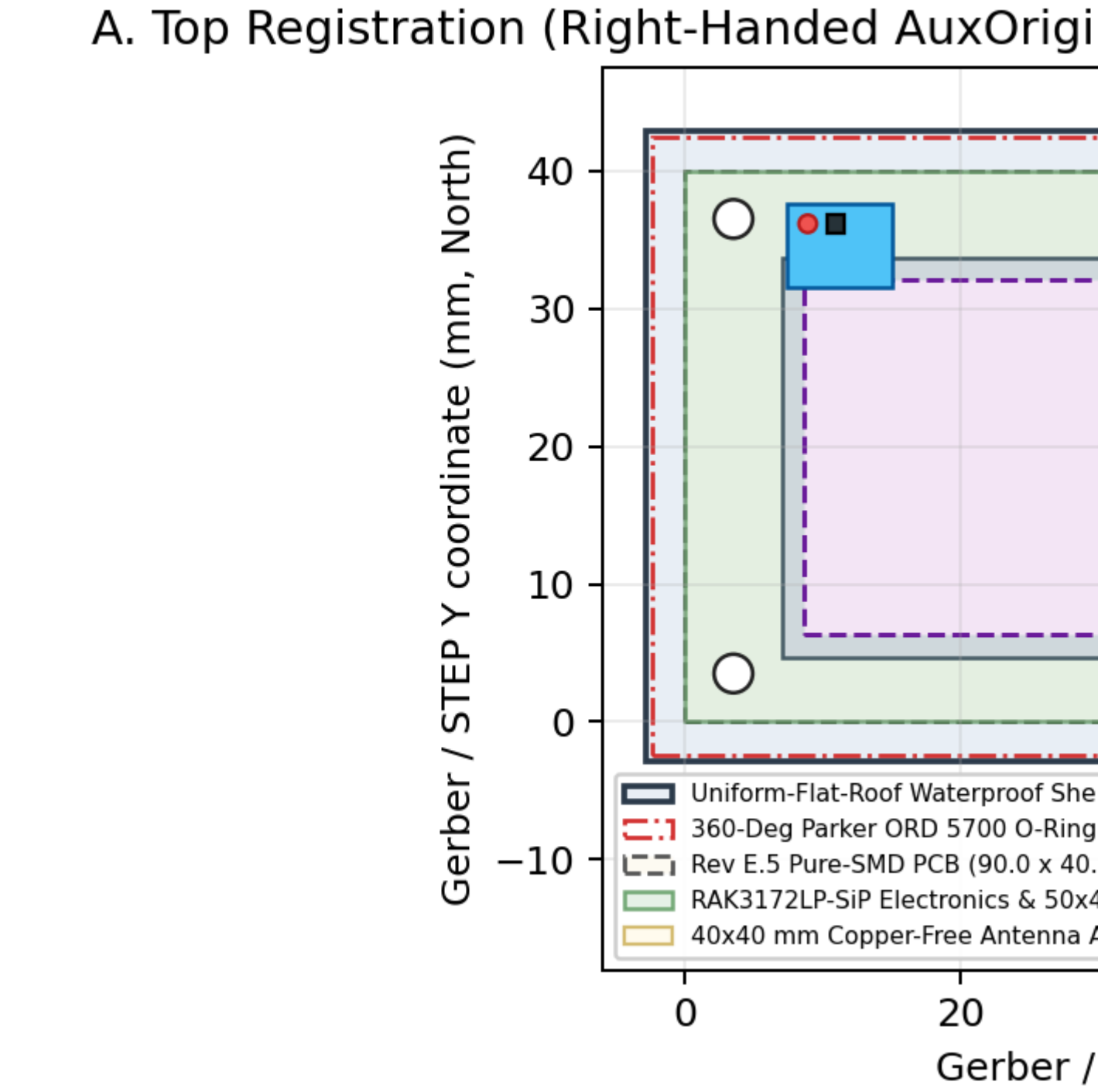


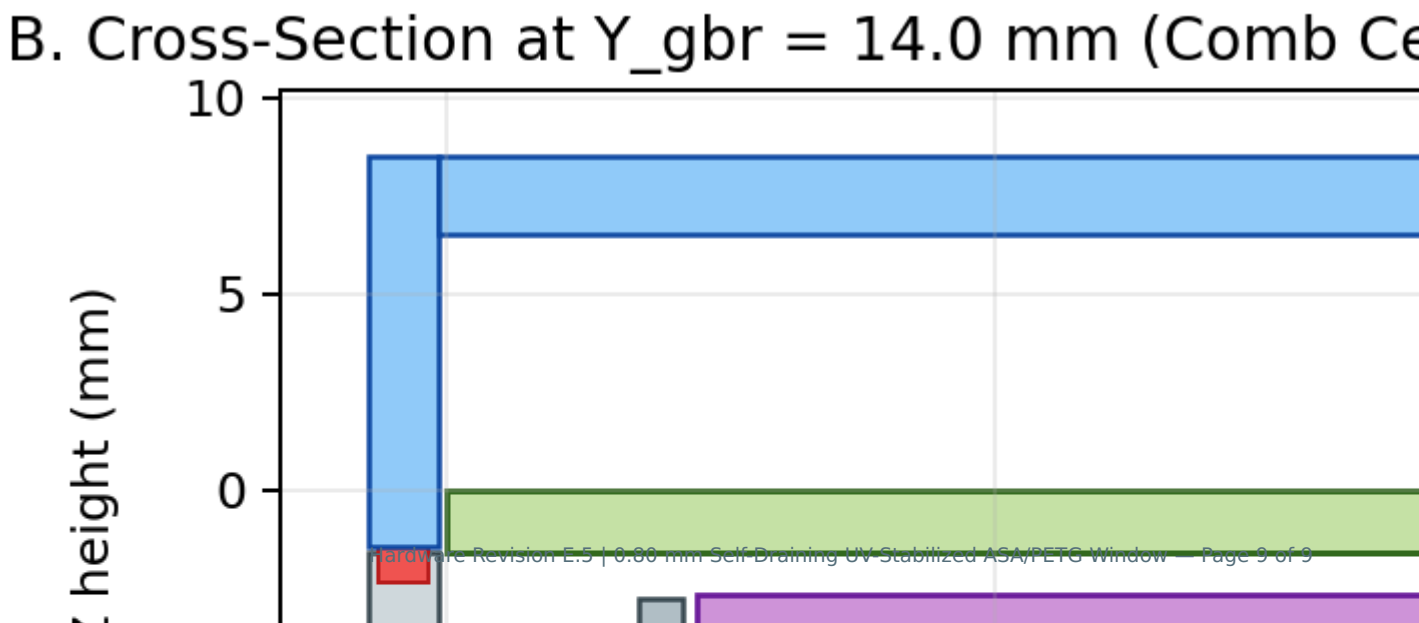
ISAC_868 Rev E.5 — Self-D

A. Exploded Assembly (Flat Lid + 3M 467MP + Pu



C. Longitudinal Cutaway @ Y_gbr = 14.0 mm (Open


A. Top Registration (Right-Handed AuxOrigi
Gerber / STEP Y coordinate (mm, North)
40
30
20
10
0
−10
0
20
Gerber /
Uniform-Flat-Roof Waterproof She
360-Deg Parker ORD 5700 O-Ring
Rev E.5 Pure-SMD PCB (90.0 x 40.
RAK3172LP-SiP Electronics & 50x4
40x40 mm Copper-Free Antenna A


B. Cross-Section at Y_gbr = 14.0 mm (Comb Ce
10
5
0
height (mm)